

$$(3) \pi_* \mathcal{M}^{\tilde{T}}(\tilde{X}) = \mathcal{M}^T(X).$$

$$(4) \pi_* \mathcal{E}^{\tilde{T}}(\tilde{X}) = \mathcal{E}^T(X).$$

Exercise 4.1.4. Show that the ergodic Bernoulli measures discussed in Example 4.5 do not exhaust all ergodic measures for the full shift as follows.

- (1) Show that any periodic orbit supports an ergodic measure which is not a Bernoulli measure.
- (2) Show that there are ergodic measures on the full shift that are neither Bernoulli nor supported on a periodic orbit.

Exercise 4.1.5. Give a different proof of Lemma 4.6 using the Radon–Nikodym derivative (Theorem A.15) and the Lebesgue decomposition theorem (Theorem A.14), instead of the pointwise ergodic theorem.

Exercise 4.1.6. Prove that the ergodic measures for the circle-doubling map $T_2 : x \mapsto 2x \pmod{1}$ are dense in the space of all invariant measures.

4.2 Ergodic Decomposition

An important consequence of the fact that $\mathcal{M}^T(X)$ is a compact convex set is that the Choquet representation theorem may be applied⁽⁴⁹⁾ to it. This generalizes the simple geometrical fact that in a finite-dimensional convex simplex, every point is a unique convex combination of the extreme points, to an infinite-dimensional result. In our setting, this gives a way to decompose any invariant measure into ergodic components.

Theorem 4.8 (Ergodic decomposition). *Let X be a compact metric space and $T : X \rightarrow X$ a continuous map. Then for any $\mu \in \mathcal{M}^T(X)$ there is a unique probability measure λ defined on the Borel subsets of the compact metric space $\mathcal{M}^T(X)$ with the properties that*

- (1) $\lambda(\mathcal{E}^T(X)) = 1$, and
- (2) $\int_X f \, d\mu = \int_{\mathcal{E}^T(X)} \left(\int_X f \, d\nu \right) d\lambda(\nu)$ for any $f \in C(X)$.

PROOF. This follows from Choquet's theorem [55] (see also the notes of Phelps [283]). A different proof will be given later (cf. p. 154), and a non-trivial example may be seen in Example 4.13. $\square$

In fact Choquet's theorem is more general than we need: in our setting, X is a compact metric space so $C(X)$ is separable, and hence $\mathcal{M}^T(X)$ is metrizable (see (B.3) for an explicit metric on $\mathcal{M}(X)$ built from a dense set of continuous functions). The picture of the space of invariant measures given by this result is similar to the familiar picture of a finite-dimensional simplex, but in fact few continuous maps⁽⁵⁰⁾ have a finite-dimensional space of invariant measures.

Indeed, as we have seen in Exercise 4.1.2, the set of ergodic measures is in general not a closed subset of the set of invariant measures.

We will see some non-trivial examples of ergodic decompositions in Sect. 4.3. The existence of the ergodic decomposition is one of the reasons that ergodicity is such a powerful tool: any property that is preserved by the integration in Theorem 4.8(2) which holds for ergodic systems holds for any measure-preserving transformation. A particularly striking case of this general principle will come up in connection with the ergodic proof of Szemerédi's theorem (see Sect. 7.2.3). There is no real topological analog of this decomposition (see Exercises 4.2.3 and 4.2.4).

Exercises for Sect. 4.2

Exercise 4.2.1. A homeomorphism $T : X \rightarrow X$ of a compact metric space (a *topological dynamical system* or *cascade*) is called *minimal* if the only non-empty closed T -invariant subset of X is X itself.

(a) Show that (X, T) is minimal if and only if the orbit of each point in X is dense.

(b) Show that (X, T) is minimal if and only if $\bigcup_{n \in \mathbb{Z}} T^n O = X$ for every non-empty open set $O \subseteq X$.

(c) Show that any topological dynamical system (X, T) has a *minimal set*: that is, a closed T -invariant set A with the property that $T : A \rightarrow A$ is minimal.

Exercise 4.2.2. Use Exercise 4.2.1(c) to prove Birkhoff's recurrence theorem⁽⁵¹⁾: every topological dynamical system (X, T) contains a point x for which there is a sequence $n_k \rightarrow \infty$ with $T^{n_k} x \rightarrow x$ as $k \rightarrow \infty$. Such a point is called *recurrent* under T .

Exercise 4.2.3. Show that in general a topological dynamical system is not a disjoint union of closed minimal subsystems.

Exercise 4.2.4. A homeomorphism $T : X \rightarrow X$ of a compact metric space is called *topologically ergodic* if every closed proper T -invariant subset of X has empty interior. Show that the following properties are equivalent:

- (X, T) is topologically ergodic;
- there is a point in X with a dense orbit;
- for any non-empty open sets O_1 and O_2 in X , there is some $n \geq 0$ for which $O_1 \cap T^n O_2 \neq \emptyset$.

Show that in general a topological dynamical system is not a disjoint union of closed topologically ergodic subsystems.